

Pavan Kumar Gummuluru

Data Scientist

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SUMMARY

Data Scientist with 3+ years of experience designing and deploying scalable ML/DL models, predictive analytics, and data-driven solutions across healthcare, IoT, and GIS domains. Proven expertise in Python, TensorFlow, PyTorch, NLP (BERT, SpaCy), and Computer Vision (CNN, GAN) for solving complex business challenges. Adept at data engineering (Snowflake, Azure Synapse, SQL) and visualization (Power BI, Tableau) to drive actionable insights. Strong background in Agile/Scrum methodologies, A/B testing, and feature engineering, with a track record of optimizing operations, reducing costs, and enhancing enterprise decision-making.

SKILLS

Programming Languages: Python, R, SQL, DAX, VBA, SAS
Methodologies: SDLC, Agile/Scrum, Waterfall, A/B Testing, Feature Engineering,
Machine Learning s Deep Learning: TensorFlow, Keras, PyTorch, Scikit-learn, CNN, RNN, LSTM, GAN
NLP: NLTK, SpaCy, BERT
Data Analysis s Processing: NumPy, Pandas, SciPy, Matplotlib, Seaborn, EDA, Data Analytics, Model Evaluation, Data Cleaning
Visualization Tools: Tableau, Power BI,
Cloud Platforms: AWS, Azure, GCP
Data Warehousing s Big Data Analytics: Azure Synapse Analytics, Snowflake, Azure Databricks
Relational Databases: MySQL, SQL, Oracle
GIS s Analytic Tools: SAP, GIS Analytics, ArcGIS, GIS, IoT, Google Analytics, Alteryx
Development s Reporting Tools: Microsoft Word, Excel, PowerPoint, Slack, Jira, Flask
Other Skills: Data Governance, Model Evaluation, Model Deployment, Problem-Solving, Statistical Analysis

EDUCATION

Masters in Data Science University of Massachusetts Dartmouth	May 2025
Bachelor in Electrical and Electronics Engineering St Martin's Engineering College	Jun 2023

EXPERIENCE

Cardinal Health, MA | Dec 2024 – Current | Data Scientist

- Developed predictive models using Python, Scikit-learn, and TensorFlow to optimize supply chain logistics, reducing operational costs by 18% through demand forecasting.
- Designed and deployed Power BI dashboards for real-time analytics, enabling stakeholders to track KPIs and make data-driven decisions.
- Performed feature engineering and A/B testing on clinical trial datasets, improving model accuracy by 22% for patient outcome predictions.
- Leveraged Azure Synapse Analytics and Snowflake to build scalable data pipelines, reducing ETL processing time by 30%.
- Collaborated with cross-functional teams to implement NLP (BERT, SpaCy) for automating medical document classification, cutting manual effort by 40%.
- Ensured data governance compliance by documenting workflows and maintaining audit trails for sensitive healthcare data.

Orion Technolab, India | Jan 2021 – July 2023 | Data Scientist

- Built deeplearningmodels (CNN, GAN) for image recognition in IoT applications, achieving 95% defect detection accuracy for manufacturing clients.
- Led an Agile/Scrum team to deliver a recommendation engine using PyTorch and AWS SageMaker, increasing client revenue by 15%.
- Optimized SQL queries and Snowflake warehouses to handle 10 TB+ datasets, improving query performance by 35% for real-time analytics.
- Developed GIS Analytics (ArcGIS) solutions to visualize spatial data for urban planning projects, enhancing decision-making for government stakeholders.
- Created custom NLP pipelines (NLTK, Transformers) to analyze customer sentiment from social media, reducing response time by 50%.
- Automated report generation using Python (Flask) and JavaScript, reducing manual effort by 20 hours/month.
- Spearheaded model evaluation frameworks to benchmark ML performance, ensuring ROI alignment for client projects.
- Mentored junior analysts in Alteryx and DAX for advanced data wrangling and Power BI dashboard development.

CERTIFICATIONS

Tableau, SQL, Power BI

PROJECT

Optimizing Image Captions with Iterative AI Feedback: A Hybrid LLM-VLM Approach | University of Massachusetts, Dartmouth

- Developed an iterative image captioning framework based on the Context-Aware Image Description (CaID) approach, integrating Large Language Models (LLMs) and Vision-Language Models (VLMs) for enhanced caption generation. Implemented task-specific prompting, iterative feedback loops, and pseudo-labeling to refine captions dynamically. Processed and analyzed the Whoops dataset (500 samples) to evaluate and improve captioning accuracy. Applied feedback-driven optimization, iteratively enhancing captions using AI-generated contextual insights. Conducted performance analysis to measure improvements across iterations and ensured a scalable, reproducible workflow using Jupyter Lab.